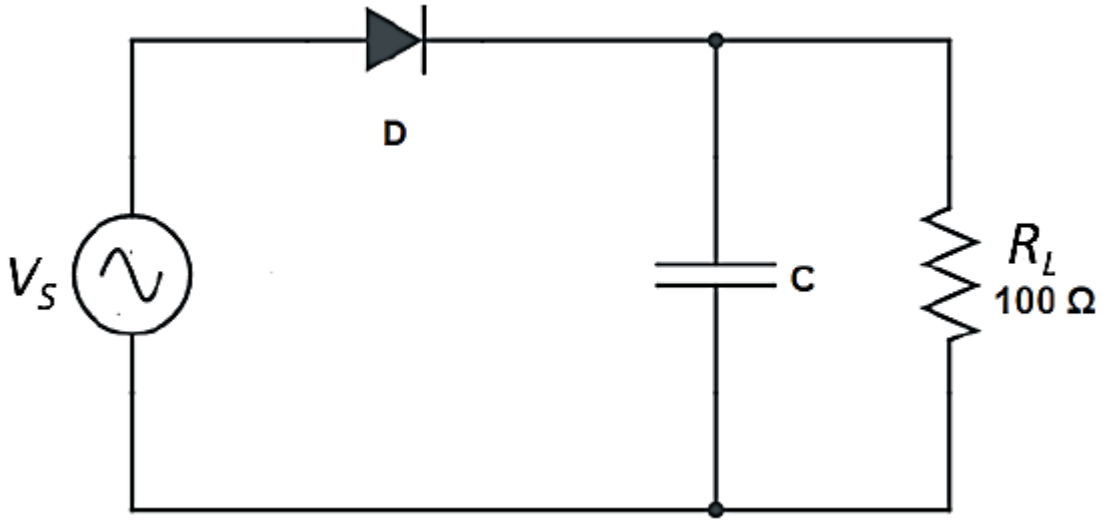
Rectifiers, Zener regulators, clippers

Practice questions · Day 05 · Wed 28 Oct · pack E3

5. Practice

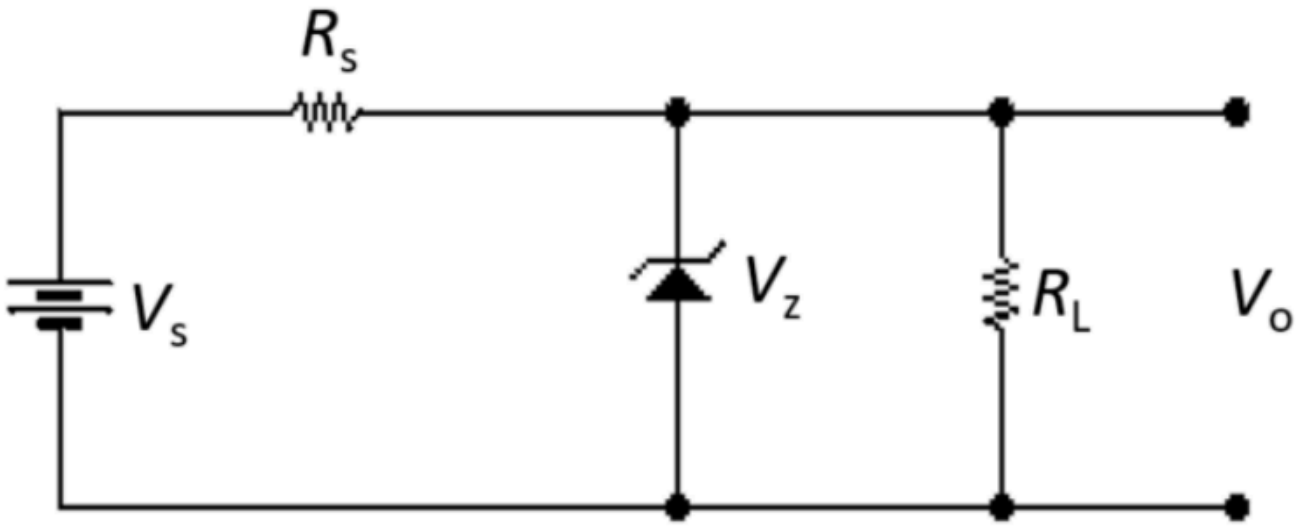
From your course files

WS3 Q9 (course question). Consider a half-wave rectifier with a filter capacitor C, where we have: $V_{S(rms)} = 12$ V, $f = 50$ Hz, $V_D = 0.8$ V, $R_L = 100\ \Omega$
a) Find the value of C which results in a ripple voltage V_{rpp} no larger than 1 V.
b) Find the following quantities: V_{Odc}: Output dc voltage, I_L: Load current, PIV: diode peak reverse voltage.



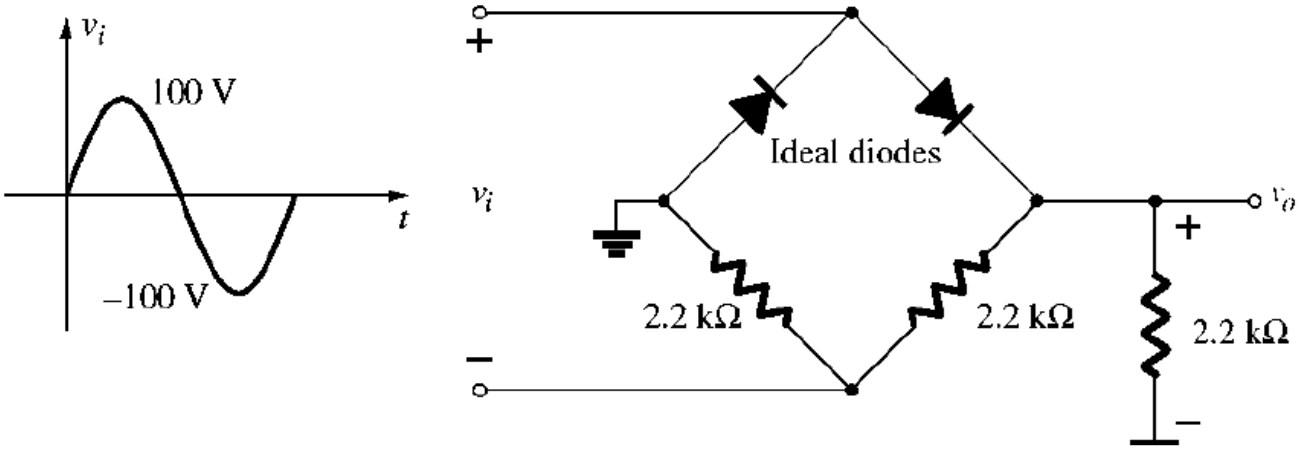
WS3 Fig. 7 (from the workshop sheet)

WS3 Q10 (course question). Consider the Zener diode circuit shows in Fig. 8 below, where: $V_{zk} = 10$ V and $r_z = 20\ \Omega$, $V_s = 15$ V, $R_s = 100\ \Omega$, and $R_L = 500\ \Omega$. Calculate V_O and I_s, then deduce I_L and I_Z. Calculate the power consumed in the Zener.



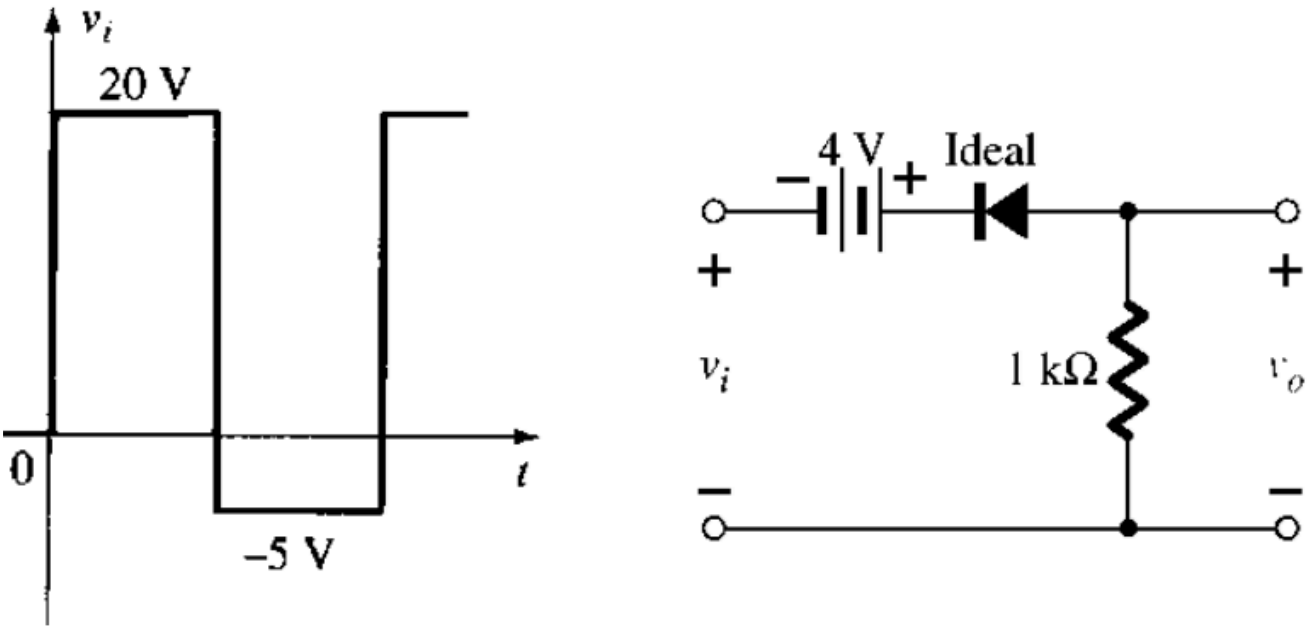
WS3 Fig. 8 (labelled “Fig. 4” on the sheet; from the workshop sheet)

WS3 Q11 (course question). Sketch v_o for the network of Fig. 9 and determine the dc voltage available.



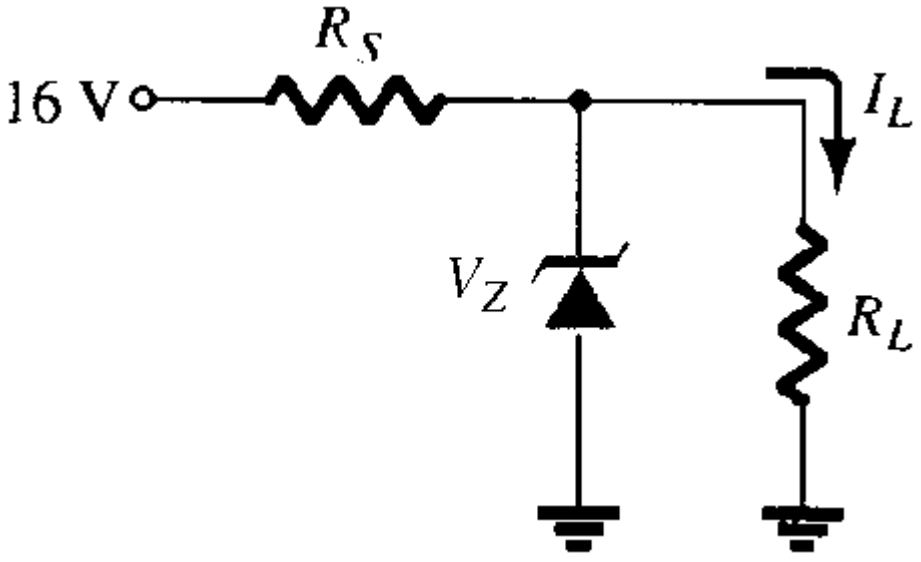
WS3 Fig. 9 (from the workshop sheet)

WS3 Q12 (course question). Determine v_o of the network of Fig. 10 for the input shown.



WS3 Fig. 10 (from the workshop sheet)

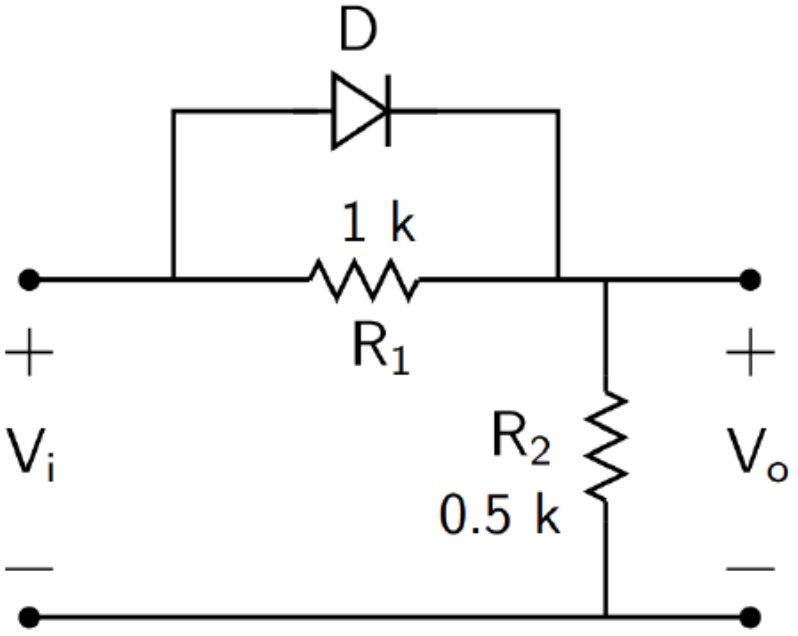
WS3 Q13 (course question). Design the network of Fig. 11 to maintain V_L at 12 V for a load variation (I_L) from 0 mA to 200 mA. That is, determine R_S and V_Z.



WS3 Fig. 11 (from the workshop sheet)

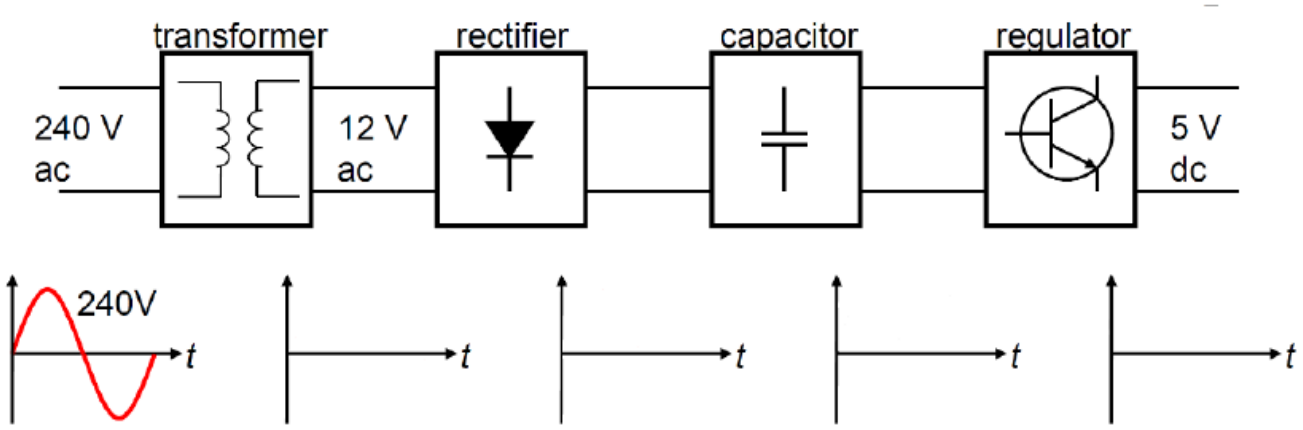
WS3 Q14 (course question). Consider a rectifier fed by a 60-Hz sinusoid having a peak value $V_P = 100$ V. Let the load resistance $R = 10$ kΩ. Find the value of the capacitance C that will result in a peak-to-peak ripple of 2 V. Also, calculate the fraction of the cycle during which the diode is conducting and the average and peak values of the diode current.

WS3 Q18 (course question). In the circuit below, the diode has a forward voltage drop $V_D = 0.7$ V. Plot V_o versus V_i for $-5 < V_i < 5$ V.



WS3 Fig. 15 (from the workshop sheet)

WS3 Q19 (course question). In the schematic below, we see the components of a typical linear DC power supply. Draw the signal at different stages of the circuit if we use half-wave or bridge rectifiers. Discuss the changes in the signal in your group. Assume that the forward voltage drop of the diode is zero.

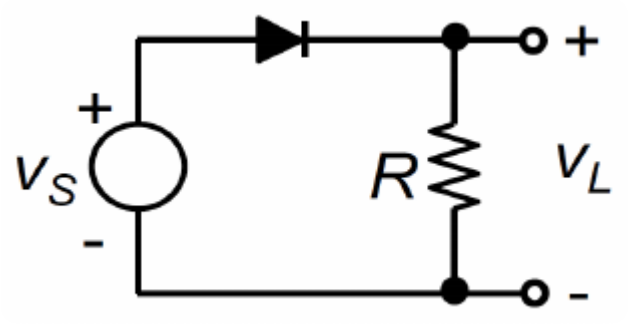


WS3 Fig. 16 (from the workshop sheet)

WS3 Q20 (course question). A diode having an internal resistance of $r_f = 20\ \Omega$ is used for half-wave rectification. If the diode forward voltage drop is zero, the supply voltage is $V_s = 50 \sin \omega t$ and load resistance is $R = 800\ \Omega$, find:

1. Maximum load current, I_m.
2. DC load current, I_{dc}.
3. Maximum voltage across the load, $V_{m,L}$.
4. DC voltage across the load, $V_{DC,L}$.
5. RMS value of the load current, I_{rms}.
6. RMS value of voltage across the load, $V_{L,rms}$.
7. DC voltage across the diode, $V_{D,DC}$.
8. Input AC power, $P_{AC,i}$.
9. Output DC power, $P_{DC,o}$.
10. The efficiency of the rectifier, η_e.

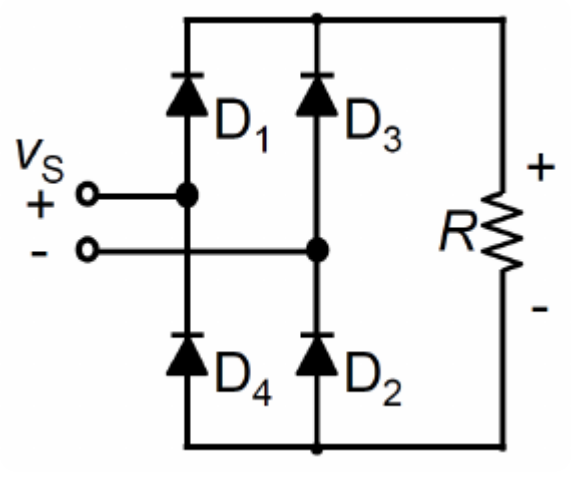
Equations that you may need: finding DC current from AC waveform, $I_{dc} = \dfrac{1}{2\pi} \displaystyle\int_0^{2\pi} I_m \sin(\omega t)\, d(\omega t)$; finding RMS current from AC waveform, $I_{RMS} = \sqrt{\dfrac{1}{2\pi} \displaystyle\int_0^{2\pi} [I_m \sin(\omega t)]^2 d(\omega t)}$.



WS3 Fig. 17 (from the workshop sheet)

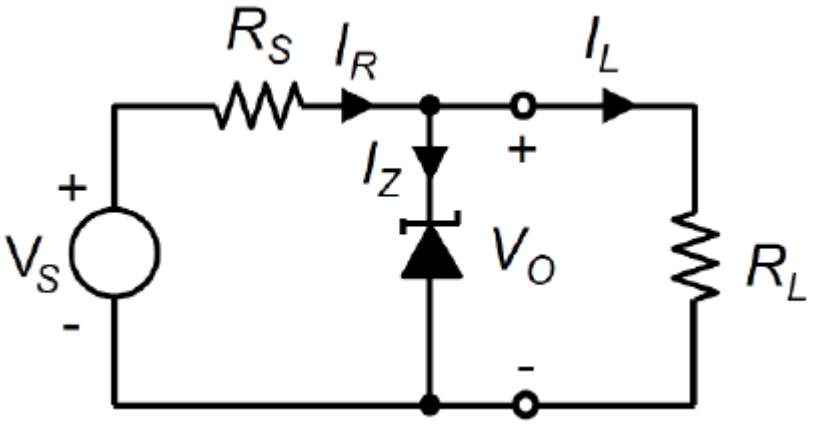
WS3 Q21 (course question). In the bridge type (full-wave rectifier) circuit shown below, the diodes have an internal resistance of $r_f = 20\ \Omega$. Assuming the diodes' forward bias voltage drop is zero, the supply voltage $V_s = 240\sin(314t)$ and the load is 460 Ω, find:

1. Maximum load current, I_m.
2. DC load current, I_{dc}.
3. Maximum voltage across the load, $V_{m,L}$.
4. DC voltage across the load, $V_{DC,L}$.
5. Peak inverse voltage (PIV) of each diode, V_{PIV}.
6. Output frequency, f_o.
7. Power dissipated by each diode, P_D.



WS3 Fig. 18 (from the workshop sheet)

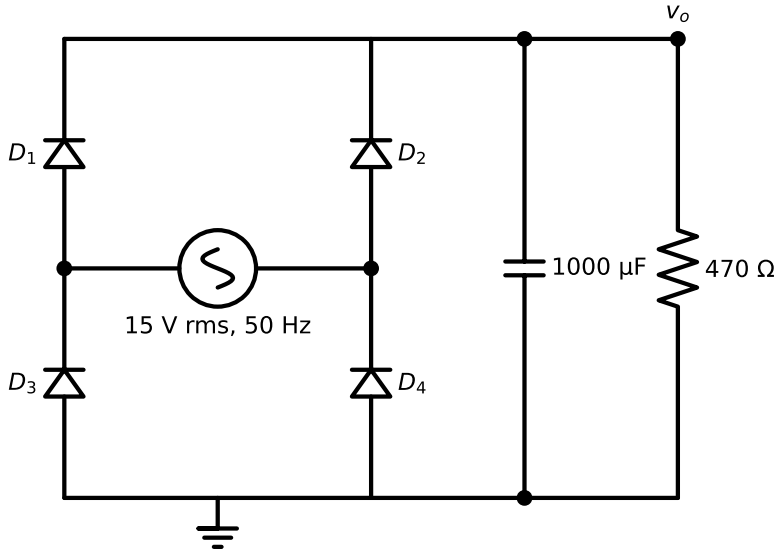
WS3 Q22 (course question). In the circuit below, we have a linear voltage regulator using a 4.8 V Zener diode. The Zener diode operates within a minimum and maximum current of $I_{Z,min} = 0.2$ mA and $I_{Z,max} = 7$ mA, respectively. Assuming that $R_s = 1$ kΩ and $V_s = 10 \pm 1$ V, obtain the maximum and minimum load, $R_{L,max}$ and $R_{L,min}$, in which the output voltage stays the same at 4.8 V.



WS3 Fig. 19 (from the workshop sheet)

New questions

N1 (easy) [4 marks]. A bridge rectifier is fed from a 15 V rms, 50 Hz secondary. The silicon diodes each drop 0.7 V. The load is 470 Ω with a 1000 µF filter capacitor. Find (a) V_P [1], (b) the peak-to-peak ripple and the ripple frequency [2], (c) $V_{O(dc)}$ and the PIV of each diode [1].

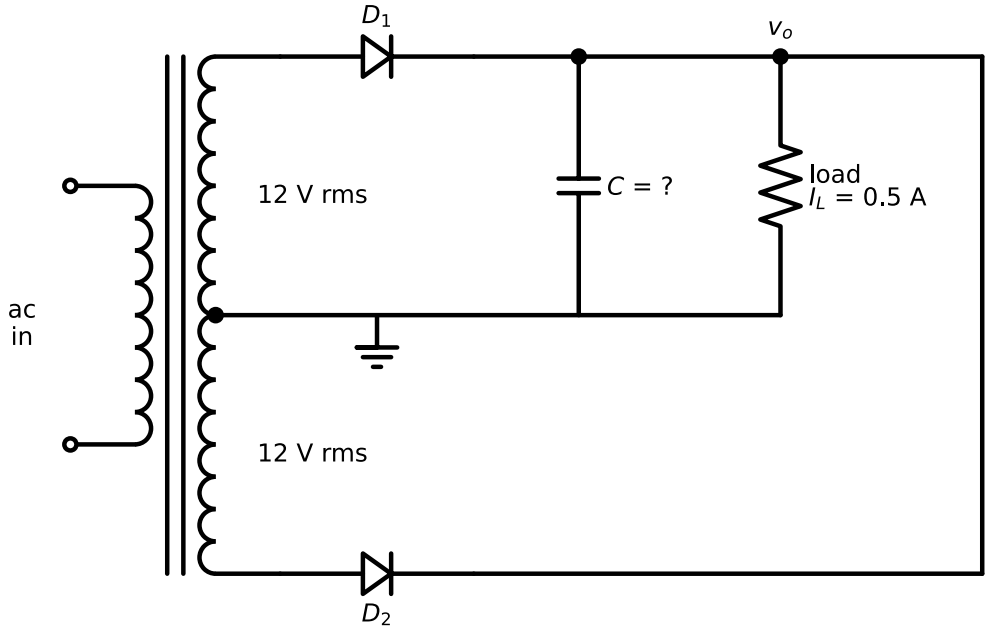


Si diodes, 0.7 V each

Circuit for N1: bridge rectifier with filter.

N2 (medium) [6 marks]. A centre-tapped full-wave rectifier has 12 V rms on each half of the secondary at 50 Hz. Diode drop is 0.7 V. It must supply 0.5 A dc with no more than 1 V peak-to-peak ripple.

a. Choose C [2]. (b) Find $V_{O(dc)}$ and the PIV of each diode [2]. (c) Find the conduction angle and the peak diode current [2].



Circuit for N2: centre-tapped full-wave rectifier.

N3 (medium) [6 marks]. A Zener with $V_{Z0} = 6.0$ V and $r_z = 5\ \Omega$ is fed from $V_s = 12$ V through $R_S = 150\ \Omega$. $R_L = 1$ kΩ.

a. Find V_O, I_Z and P_Z [3]. (b) V_s rises to 13 V. Find the new V_O and the line regulation in mV/V [2]. (c) With $V_s = 12$ V, find the % regulation between no load and $R_L = 1$ kΩ [1].

Circuit for N3: the Zener drawn as its model, V_{Z0} in series with r_z.

N4 (hard, chained) [10 marks]. Design a 9.1 V supply for a load drawing 0–50 mA. Use a bridge rectifier from an 18 V rms, 50 Hz secondary (Si diodes, 0.7 V each), a filter capacitor allowing at most 2 V peak-to-peak ripple, and a 9.1 V Zener that needs $I_{Z,min} = 5$ mA.

a. Find V_P and the lowest voltage the capacitor reaches [2].
b. Find the largest R_S that keeps regulation, and choose a standard value [2].
c. Explain why I_S does not depend on the load. Using the average capacitor voltage, find I_S and choose C [3].
d. Find the worst-case Zener power and the diode PIV [2].
e. Estimate the peak diode current [1].

Circuit for N4: bridge, filter capacitor, series resistor and 9.1 V Zener.

N5 (medium) [5 marks]. A sine wave $v_i = 10 \sin \omega t$ V drives a 1 kΩ series resistor. v_o is taken across two parallel branches to ground. Branch 1 is diode D_1 (anode at the output) in series with a 3 V battery whose + terminal is at the cathode of D_1 and whose − terminal is at ground, so the cathode sits at +3 V. Branch 2 is diode D_2 (cathode at the output) in series with a 2 V battery whose − terminal is at the anode of D_2 and whose + terminal is at ground, so the anode sits at −2 V. Both diodes are Si, 0.7 V.

a. Find the transition voltages and write v_o in each region [3]. (b) Sketch v_o and find the peak current in the 1 kΩ [2].

Circuit for N5: double clipper.

N6 (easy) [4 marks]. (a) A clamper (series C, ideal diode from output to ground with its anode at ground) is driven by a ±10 V square wave. Give the output levels [2]. (b) A half-wave voltage

doubler runs from 12 V rms. Find the ideal output, the output with 0.7 V diodes, and the PIV of each diode [2].

Circuits for N6: (a) clamper, (b) half-wave voltage doubler.

6. Quick recall check

1. A bridge rectifier and a half-wave rectifier use the same C, R and V_P. What is the ratio of their ripple voltages?
2. Write the formula for C that gives a peak-to-peak ripple V_r in a half-wave supply with load current I_L.
3. In a Zener shunt regulator, at which load and supply conditions do you size R_S, and at which do you check P_Z?
4. What is the PIV of each diode in a centre-tapped rectifier with V_S peak per half-winding?
5. What happens to a varactor's capacitance when the reverse bias increases, and why?